

GIRIDHARAN S

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About Me

3rd-year Electronics Engineering (VLSI) student with hands-on experience in DFT, RTL design, Physical Design, and CMOS VLSI using industry-standard EDA tools. Skilled in Verilog, Siemens Tessent, VCS, Verdi, and Fusion Compiler, with practical exposure to RTL-to-OASIS implementation and RISC-V. Seeking a semiconductor internship to apply my skills, learn from industry professionals, and contribute to real-world silicon development.

Education

2024 – 2028
Chennai, India **B.E - Electronics Engineering (VLSI Design & Technology),**
Rajalakshmi Institute of Technology | CGPA: 8.00/10

Experience

Jul 2026 – Present
Chennai, India **Rajalakshmi Institute of Technology, Hands-on DFT Assignment — Siemens Tessent**

- Independently set up Siemens Tessent in a Linux environment, including installation, environment configuration, Tessent Shell setup, and design setup.
- Performed hands-on DFT operations involving scan and scan-chain concepts and ATPG-related workflows, developing practical understanding of the Tessent DFT flow.
- Debugged tool, environment, and design-setup issues while working through DFT operations and building hands-on experience with DFT methodology.

Nov 2025 – Jan 2026
Remote **Infosys Springboard, Technical Intern – Python**

- Completed a 2-month virtual internship focused on applied Python, delivering project-based solutions and hands-on exercises while applying OOP, data structures, and file I/O concepts.
- Developed Python scripts for task automation and data processing, using systematic testing and debugging to improve code reliability and problem-solving skills.
- Completed coding challenges and real-world exercises, documented solutions, and presented work to mentors to strengthen technical communication and software development practices.

Projects

06/2026–07/2026 **RTL ALU — Design, Verification & RTL-to-OASIS** Designed and implemented an ALU using Verilog RTL and validated its functionality through Synopsys VCS simulation and Verdi waveform analysis. Implemented the design through Synopsys Fusion Compiler, including floorplanning, power planning, placement, CTS/clock optimization, routing, and OASIS generation.

07/2026-08/2026 **RISC-V Processor — RTL Debugging, Verification & Physical Implementation** Analyzed the provided RISC-V processor RTL architecture and developed a self-written Verilog testbench for functional validation. Used Synopsys VCS and Verdi for simulation and waveform debugging, followed by physical implementation through Fusion Compiler toward OASIS output.

2026 – Present **Adaptive Multi-Stage Hardware Recovery Architecture for Performance- and Reliability-Aware Self-Healing RISC-V SoC** Academic project focused on performance- and reliability-aware hardware recovery architecture for a RISC-V SoC.

10/2025-11/2025 **Air Mouse — Gesture-Controlled Human Interface** Developed a wireless air-mouse system for cursor control using hand gestures. Integrated motion sensing and controller-based processing to enable real-time gesture-based computer interaction.

04/2025 -08/2025 **Vision Speak — AI-Powered Assistive System for Visually Impaired People** Developed an ESP32-CAM-based assistive system integrating obstacle detection, OCR-based text reading, object/currency identification, and face recognition. Provided audio and vibration feedback through a compact wearable design to support safer mobility and greater independence.

Academic **Custom CMOS VLSI Design & Layout** Designed transistor-level CMOS schematics and layouts using Synopsys Custom Compiler and Cadence Virtuoso. Worked on latches, flip-flops, shift registers, counters, SRAM, register files, adders, and multipliers, including DRC, LVS, and PEX verification.

Technical Skills & Achievements

Programming: C | Java | Python
Computer Architecture: RISC-V ISA | Computer Architecture | Digital Logic
HDL & RTL Design: Verilog HDL | SystemVerilog (Learning) | RTL Design | RTL Simulation & Debugging
FPGA: Altera FPGA | Intel Quartus | ModelSim | Synthesis | Implementation
Physical Design: RTL-to-OASIS | Floorplanning | Power Planning | Placement | CTS | Routing | Timing & Power Analysis
DFT: Siemens Tessent | Tessent Shell | Scan/Scan Chain | DFT Operations | ATPG-related Workflows | DFT Debugging
EDA Tools: Fusion Compiler | VCS | Verdi | Custom Compiler | Design Compiler (Learning) | Cadence Virtuoso
OS: Linux

Certifications & Training
ProV Logic — RTL-to-GDSII Workshop (June 13–14, 2026)
Infosys Springboard — Python Virtual Internship
Siemens Tessent — DFT Training/Certification
Maven Silicon — VLSI System-on-Chip (SoC) Course

Hackathon & Achievements
Smart India Hackathon — Hardware Connections & Presentation Contribution
INNOCLASH'26 — Hackathon